

# Algorithm Design Paradigms

## Handout on Induction

27 January 2026

*The rise of nonchalance is the death of passion*

— Anonymous

### §1 Induction

**Principle of mathematical induction 1.1.** Suppose we have an infinite sequence of statements

$\varphi_0, \varphi_1, \varphi_2, \varphi_3, \dots$

and we can prove the following 2 statements -

- $\varphi_0$  is true
- For each  $n > 0$ , if  $\varphi_{n-1}$  is true, then  $\varphi_n$  is also true.

then all the statements  $\varphi_0, \varphi_1, \varphi_2, \varphi_3, \dots$  in the sequence are true.

The above definition should be read as follows, given a sequence of formulas:

- The first one is true.
- Any formula being true, implies that the next one in the sequence is true.

Then all of the formulas in the sequence are true. Something like a chain of dominoes falling.

#### ☰ Exercise

Show that  $n^2$  is the same as the sum of first  $n$  odd numbers using induction.

#### ☰ The scenic way

(a) Prove the following theorem of Nicomachus by induction:

$$\begin{aligned}1^3 &= 1 \\2^3 &= 3 + 5 \\3^3 &= 7 + 9 + 11 \\4^3 &= 13 + 15 + 17 + 19 \\&\vdots\end{aligned}$$

(b) Use this result to prove the remarkable formula

$$1^3 + 2^3 + \dots + n^3 = (1 + 2 + \dots + n)^2$$

### ☰ Same Height?

Here is a proof by induction that all people have the same height. We prove that for any positive integer  $n$ , any group of  $n$  people all have the same height. This is clearly true for  $n = 1$ . Now assume it for  $n$ , and suppose we have a group of  $n + 1$  persons, say  $P_1, P_2, \dots, P_{n+1}$ . By the induction hypothesis, the  $n$  people  $P_1, P_2, \dots, P_n$  all have the same height. Similarly the  $n$  people  $P_2, P_3, \dots, P_{n+1}$  all have the same height. Both groups of people contain  $P_2, P_3, \dots, P_n$ , so  $P_1$  and  $P_{n+1}$  have the same height as  $P_2, P_3, \dots, P_n$ . Thus all of  $P_1, P_2, \dots, P_{n+1}$  have the same height. Hence by induction, for any  $n$  any group of  $n$  people have the same height. Letting  $n$  be the total number of people in the world, we conclude that all people have the same height. Is there a flaw in this argument?

### ☰ There is enough information!

Given  $a_0 = 100$  and  $a_n = -a_{n-1} - a_{n-2}$ , what is  $a_{2025}$ ?

### ☰ 2-3 Color Theorem

A  $k$ -coloring is said to exist if the regions the plane is divided off in can be colored with  $k$  colors in such a way that no two regions sharing some length of border are the same color.

- (a) A finite number of circles (possibly intersecting and touching) are drawn on a paper. Prove that a valid 2-coloring of the regions divided off by the circles exists.
- (b) A circle and a chord of that circle are drawn in a plane. Then a second circle and chord of that circle are added. Repeating this process, until there are  $n$  circles with chords drawn, prove that a valid 3-coloring of the regions in the plane divided off by the circles and chords exists.

### ☰ Square-full

Call an integer square-full if each of its prime factors occurs to a second power (at least). Prove that there are infinitely many pairs of consecutive square-fulls.

Hint: We recommend using induction. Given  $(a, a + 1)$  are square-full, can we generate another?

### ☰ Proving the principle of induction

Prove that the following statements are equivalent -

- every nonempty subset of  $\mathbb{N}$  has a smallest element
- the definition of induction

You can assume that  $<$  is a linear order on  $\mathbb{N}$  with  $n - 1 < n$  and such that there are no elements strictly between  $n - 1$  and  $n$ .